MILITARY
SPECIFICATION

MIL-P-12965(SigC)
Amendment No. 3
19 May 1955

POWER SUPPLY (RECTIFIER POWER UNIT RA-61-( )) AND
MOUNTING FT-414-( )

1. Paragraph 2.1

a. Delete the following specifications:

FEDERAL

NN-B-591	Boxes, Fiberboard, Wood-Cleated (For Domestic Shipment).
NN-B-601	Boxes, Wood-Cleated-Plywood, For Domestic Shipment.
LLL-B-631	Boxes, Fiber, Corrugated (For Domestic Shipment).

MILITARY

JAN-P-105	Packaging and Packing For Overseas Shipment-Boxes, Wood-Cleated, Plywood.
MIL-T-10513	Tropicalization of Material Used in Signal Corps Equipment.

U. S. ARMY

19-10	Filler, Interior Packing, Fiberboard, Corrugated, Signal and Double-Faced, for General Packing Purposes.
71-4944	Identification of Ground Signal Equipment.

b. Delete the following drawings:

460-12	Group B-2 Sampling Plans.
SC-D-6642	Mounting FT-414.

c. Add the following specifications:

FEDERAL

NN-B-631	Boxes; Wood, Wirebound (For Domestic Shipment).
PPP-B-601	Boxes; Wood, Cleated, Plywood.

MILITARY

MIL-P-117	Packaging and Packing for Overseas Shipment - Bags, Interior Packaging.
MIL-M-13231	Marking of Electronic Items.

Source: https://assist.dla.mil -- Downloaded: 2026-04-17T23:54Z
Check the source to verify that this is the current version before use.

d. Add the following standards:

MIL-STD	Wired Equipment, Classification of Visual and Mechanical Defects For.

e. Add the following drawings:

SC-DL-87773	Mounting FT-414( ).

2. Insert the following paragraph:

2.2 Other publications.- The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

ASSOCIATION OF AMERICAN RAILROADS PUBLICATION

Uniform Freight Classification Rules and Container Specifications for Rail Shipments.

(Application for copies of the above publication should be addressed to Association of American Railroads, Freight Loading and Container Bureau, 59 Van Buren Street, Chicago 5, Illinois.)

AMERICAN TRUCKING ASSOCIATION PUBLICATION

National Motor Freight Classification Rules and Container Specifications For Truck Shipments.

(Application for copies of the above publication should be addressed to the American Trucking Association, 1424, 16th Street N.W., Washington D. C.)

3. Paragraph 3.2 Procurement Model.- Insert "(Power Supply)" after "The Procurement Model."

4. Paragraph 3.2.3 Performance.- In first sentence, replace "Equipment" with "Power Supply."

5. Paragraph 3.4.4 Permanency and Legibility.- In first sentence, replace "71-4944" with "MIL-M-13231."

6. Paragraph 3.7 Tropicalization.- Delete second sentence.

7. Paragraphs 3.9.1 and 3.9.2 - Replace with the following:

3.9.1 Equipment Operating.- After a minimum warm-up period, the equipment shall meet the performance requirements of this specification while subjected to any of the following conditions successively or in combinations likely to be encountered during worldwide operation:

Source: https://assist.dla.mil -- Downloaded: 20260-17T23:54Z
Check the source to verify that this is the current version before use.

a. Operation.- Continuous operation for a period of 1 year, alternating 4 hours on and 2 hours off, with no more than normal maintenance and replacement of parts. Compliance will be determined by examination of the preproduction samples (See 3.3).

b. Temperature.- Ambient temperature in the range of +140°F to -40°F. Exposure at the high temperature extreme not to exceed 4 hours and at the low temperature extreme not to exceed 72 hours. (See 4.4).

c. Relative humidity.- Relative humidity up to 97 percent; and exposure at 100 percent relative humidity, with condensation, for 4 hours. (See 4.5).

3.9.2 Equipment Nonoperating.- The equipment shall comply with the operational requirements of 3.9.1 after subjection to any of the following nonoperating conditions successively or in combinations encountered during worldwide short-time storage and transit:

a. Temperature.- Continuous exposure in the range of +160°F to -80°F, however, exposure at the high temperature extreme not to exceed 4 hours and at the low temperature extreme not to exceed 24 hours.

b. Relative humidity.- Relative humidity up to 97 percent, for an indefinite period of time; and exposure at 100 percent relative humidity, with condensation, for 4 hours. Following subjection to such conditions and and a power-on period of 16 hours at normal room temperature and relative humidity, the equipment shall meet the requirements of 3.9.1 (See 4.5).

c. Elevation.- Elevation up to 25,000 feet above sea level.

d. Orientation.- Indefinite storage in any orientation.

e. Vibration, bounce and shock.- As described in 3.20 and 3.22.

8. Paragraph 3.14: Case.- Delete last two sentences.

9. Paragraph 3.19: Replace with the following:

3.19 Mounting FT-414.- Mounting FT-414 shall be constructed in accordance with Drawing SC-DL-87773.

10. Paragraph 4.1.1 Group A Inspection.- Replace Table II with the following:

Table II.- Government group A inspection

Inspection	Req Par	Test Par	AQL
Power supply unit			
Input and output voltage	3.11	4.6	1.5 dphu*
Visual and mechanical, major		4.9	1.5 dphu
Visual and mechanical, minor		4.9	6.5 dphu
Visual and mechanical, control		4.9	10%
Dimensions		4.10	0.65%
Mounting FT-414			
Visual and mechanical, major		4.9	1%
Visual and mechanical, minor		4.9	4%
Visual and mechanical, control		4.9	10%
Dimensions		4.10	0.65%

* Defects per hundred units

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:54Z
Check the source to verify that this is the current version before use.

Amend No. 3 to
Spec MIL-P-12965(SigC)

11. Paragraph 4.1.2 Group B Inspection.- Replace Table II with the following:

Table III.- Government group B inspection

Inspection	Req Par	Test Par	Group B Plan
Ripple Voltage	3.12	4.7	) B-2 combined
Voltage Regulation	3.13	4.8	) (See Table IV)

12. Following Table II, insert:

Table IVa - Group B-2 sampling plans

Lot size /&	Reduced Inspection /#		Normal Inspection /#		Tightened Inspection /#	
	Sample size	Acceptance number	Sample size	Acceptance number	Sample size	Acceptance number
9 to 25, incl	2	0	2	0	4	0
26 to 65, incl	2	0	3	0	6	0
66 to 180, incl	7	1	7	1	17	1
181 to 500, incl	7	1	17	2	28	2
501 to 1,300, incl	7	1	28	3	41	3
1,301 to 3,200, incl	17	2	41	4	53	4
3,201 to 8,000, incl	17	2	53	5	66	5
8,001 to 22,000, incl	28	3	66	6	81	6
22,001 and over	28	3	81	7	81	6

/& For lot sizes under 9, the sample size shall be at the option of the Government but not more than 2, and the acceptance number shall be 0.

/# Normal, tightened, or reduced inspection shall be used in accordance with the following: (Part "b" of this table lists values for limits, A, B, and C, referenced below.)

(1) Calculate the estimated process average from the formula 100 P/N; where P equals the number of defective units found in N sample units, and N equals the number of sample units included in the estimated process average. The Government will determine the number of consecutive sample units to be used to estimate the process average.

(2) Except as stated in (3) below, use normal inspection as long as the process average is between limits A and C.

(3) Change to tightened inspection whenever the process average becomes equal to or greater than limit A, or whenever a lot fails to pass.

(4) Return to normal inspection from tightened inspection when two lots in succession pass tightened inspection, and the process average becomes less than limit B.

(5) Change to reduced inspection whenever the process average becomes less than limit C.

(6) Return to normal inspection from reduced inspection when the process average becomes equal to or greater than limit C.

Document has been created by Document Assistant on 2024-10-17T23:54Z
Check the source to verify that this is the current version before use.

Table IVb.- Group B-2 limits of the process average

N	Limit A %	Limit B %	Limit C %	N	Limit A %	Limit B %	Limit C %
10	18.78	11.89	*	130-144	8.82	6.91	1.18
11	18.14	11.57	*	145-159	8.62	6.81	1.38
12	17.58	11.29	*	160-179	8.45	6.72	1.55
13	17.09	11.04	*	180-199	8.25	6.62	1.75
14	16.65	10.82	*	200-249	8.08	6.54	1.92
15-16	16.25	10.63	*	250-299	7.76	6.38	2.24
17-18	15.57	10.29	*	300-349	7.52	6.26	2.48
19-20	15.00	10.00	*	350-399	7.33	6.16	2.67
21.23	14.51	9.76	*	400-449	7.18	6.09	2.82
24.26	13.90	9.45	*	450-499	7.05	6.03	2.95
27-29	13.39	9.19	*	500-549	6.95	5.97	3.05
30-34	12.96	8.98	*	550-599	6.86	5.93	3.14
35-39	12.37	8.68	*	600-649	6.78	5.90	3.22
40-44	11.89	8.45	*	650-699	6.71	5.85	3.29
45-49	11.50	8.25	*	700-749	6.65	4.82	3.35
50-59	11.16	8.08	*	750-799	6.59	5.80	3.41
60-69	10.63	7.81	*	800-849	6.54	5.77	3.46
70-79	10.21	7.60	*	850-899	6.50	5.75	3.50
80-89	9.87	7.44	0.13	900-949	6.45	5.73	3.55
90-99	9.59	7.30	0.41	950-999	6.41	5.71	3.59
100-114	9.36	7.18	0.64	1000	6.38	5.69	3.62
115-129	9.06	7.03	0.94				

* Number of sample units included in estimated process average is insufficient for reduced inspection.

N = Number of sample units in estimated process average.

13. Paragraph 4.1.2.2: Replace with the following:

4.1.2.2 Sampling procedure.- Sampling shall be conducted in accordance with Table III. Samples shall be selected, without regard to their quality, by the Government Inspector. Unless otherwise specified, normal inspection shall be used at the start of the contract. If the number of defective sample units from a lot exceeds the acceptance number specified in the acceptance plan, the lot shall be rejected. Disposition of the rejected product (Sample Unit and Lots) shall be in accordance with Standard MIL-STD-105 and 4.1.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:54Z
Check the source to verify that this is the current version before use.

14. Paragraph 4.4.1 Low Temperature Test.- In first sentence, change "24" to "72". In third sentence, change "72" to "24".

15. Paragraph 4.4.2 High Temperature Test.- In first sentence change "At least 4 hours" to "Not more than 4 hours." In second and third sentences, change "+125°F" to "+140°F."

16. Paragraph 4.9: Replace with the following:

4.9 Visual and mechanical inspection.- Parts and equipment shall be examined for the defects listed in Standard MIL-STD-252.

17. Add the following paragraph:

4.10 Dimensional inspection.- Selected dimensions shall be gaged or measured to determine compliance with this specification. The dimensions to be checked shall be selected at a meeting between the contractor and the contracting officer, held as soon as practical after the manufacturer's shop drawings have been prepared but before equipment is submitted for acceptance.

18. Replace Section 5 with the following:

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level A - Military package.- Power Supply (Rectifier Power Unit RA-61( )) and Mounting FT-414( ) shall be preserved and packaged in accordance with the procedures specified for the designated methods as prescribed in Specification MIL-P-116, and further described in the following paragraphs.

5.1.1.1 Technical literature.- Package each Technical Literature Method IC-3, as follows: Inclose each technical literature within a close-fitting, Class b bag conforming to Specification JAN-P-117. Properly seal closure.

5.1.1.2 Tubes.- No packaging required. Mount the spare tube(s) in the dummy socket on the chassis of the unit.

5.1.13 Fuses.- Place the spare fuses within the mountings provided on the chassis of the unit.

Source: https://assist.dla.mil -- Downloaded: 202610-17T23:54Z
Check the source to verify that this is the currentversion before use.

5.1.1.4 Power Supply (Rectifier Power Unit RA-61( )).- Package each power supply Method IIb, as follows: Remove the cords from the power supply. Wind each cord into a coil of proportionate dimensions. Tie at 3 places with cotton tape, securing the terminals to the coil with the ties. Secure cords to the unit. Cushion each power supply on all surfaces with cells or pads or both fabricated of corrugated fiberboard 200 pound Mullen Test strength and designed to absorb the shock of impact normally encountered in handling and transit. Place each cushioned power supply together with the required amount of desiccant as prescribed in Specification MIL-P-116 within a close-fitting, suitable style and grade corrugated fiberboard box. Seal the box closure with gummed paper tape conforming to Specification UU-T-111. Blunt all corners of the box. Place the boxed power supply within a water-vapor-proof barrier qualified for Class 1 application in accordance with the requirements of Specification MIL-B-131. Heat seal closure. Place the water-vapor-proofed power supply within a close-fitting, suitable style and grade fiberboard box conforming to the requirements of Specification JAN-P-108. Seal the entire box closure, corners and joints with water-resistant, pressure-sensitive tape conforming to Specification JAN-P-127.

5.1.1.5 Mounting FT-414( ).- Package each mounting Method IC-5, as follows: Cushion each mounting by wrapping in flexible corrugated paper. Secure the cushioning with gummed paper tape. Place the cushioned mounting with a close-fitting fiberboard box, conforming to the requirements of Specification JAN-P-108. Seal the entire box closure, corners and joints with water-resistant, pressure-sensitive tape Specification JAN-P-127.

5.1.2 Level B - Limited military package.-

5.1.2.1 Technical literature shall be packaged as specified in 5.1.1.1.

5.1.2.2 Tubes shall be packaged as specified in 5.1.1.2.

5.1.2.3 Fuses shall be packaged as specified in 5.1.1.3.

5.1.2.4 Power Supply (Rectifier Power Unit RA-61( )), shall be packaged as specified in 5.1.1.4.

5.1.2.5 Mounting FT-414( ), shall be packaged as specified in 5.1.1.5.

5.1.3 Level C - Minimum military package.- Power Supply (Rectifier Power Unit RA-61( )) and Mounting FT-414( ) shall be packaged in accordance with commercial practice and in such a manner that will guarantee adequate protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity at the lowest transportation rate.

5.2 Packing.-

sa

Source: https://assist.dla.mil -- Downloaded: 202610-17T23:54Z
Check the source to verify that this is the current version before use.

5.2.1 Level A - military pack.- Power Supply (Rectifier Power Unit RA-61( )) and Mounting FT-414( ), shall be packed individually within wood-cleated plywood, nailed wood or wirebound boxes conforming to the requirements of Specifications PPP-B-601, JAN-P-106 or MIL-B-107 respectively. Fabricate the box to fit the contents snugly. Shipping containers shall be closed as specified in the appendix of the applicable box specification.

5.2.1.1 Shipping containers shall be strapped in conformance with the requirements of the appendix of the applicable box specification only for direct shipment to ports.

5.2.2 Level B - Limited military pack.- Power Supply (Rectifier Power Unit RA-61( )) and Mounting FT-414( ), shall be individually packed within wood cleated plywood, nailed wood or wirebound boxes, conforming to the requirements of Specifications PPP-B-601, NN-B-621 or NN-B-631 respectively. Fabricate the box to fit the contents snugly. Close the shipping containers as specified in the appendix of the applicable box specification.

5.2.3 Level C - Minimum military pack.- Power Supply (Rectifier Power Unit RA-61( )) and Mounting FT-414( ) shall be packed for shipment in conformance with the requirements of Uniform Freight Classification Rules and Container Specification for rail shipment. National Motor Freight Classification Rules and Container Specification for truck shipment, Parcel Post Regulations and the regulations of other carriers as applicable to the mode of transportation.

5.3 Marking for shipment and storage.- Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 202610-17T23:54Z
Check the source to verify that this is the current version before use.